ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) (i) $f_x = f/p = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.

(ii) Sensor width = 1280 × 5 µm = 6.4 mm. Using similar triangles, FOV = $W_s \cdot f / Z = 6.4 \times 16/400 =$ **0.256 mm**.

Distance per pixel = FOV/1280 = 0.256/1280 = $\mathbf{2\times10^{-4}}$ **mm** = 0.2 µm.

(b) Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisaton sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5$ V/4096 = 1.2207×10^{-3} V = 1.2207 mV.

The sensor maps force to voltage by V = S·F with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S$ = 1.2207 mV / (10 mV/N) = 0.12207 N. $\mathbf{\Delta F_{min} \approx 0.122\ N}$. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) $F = e^{AT}$, $G = A^{-1}(F - I)B$. Taking T = 0.01 s.

$F = e^{-0.02} =$ **0.9802**.

G = (0.9802 − 1)/(−2) = **0.0099**.

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) (1) Diferentiation amplifies noise in the encoder signal.

(2) Finite differencing needs a lot of memory and processing, wich is slow on a micro-controler at 1 kHz.

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20$ = 125.66 rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s$ = 0.001 s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a) z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a) z^{-1}]$.

Dividing through by (1 + a): a/(1 + a) = 0.062832/1.062832 = 0.059117 and (1 − a)/(1 + a) = 0.937168/1.062832 = 0.881767. Hence with u = differenced signal and y = filtered velocity: $\mathbf{y_k = 0.05912\ (u_k + u_{k-1}) + 0.88177\ y_{k-1}}$. (Frequency pre-warping is neglected sence $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant (z = 1): y = 2(0.05912)u + 0.88177 y ⇒ y(1 − 0.88177) = 0.11824 u ⇒ y/u = 0.11824/0.11823 = 1.000. In general H(1) = 2a/[(1 + a) − (1 − a)] = 2a/2a = 1 exactly, so the steady-state output equals the commanded input ✓.

(c) A − LC = $[-\ell_1, 1; -\ell_2, -2]$; characteristic polynomial $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desiered $(s + 10)^2 = s^2 + 20s + 100$. Then $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$, and $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 18 = 82$.

$\mathbf{L = [18, 82]^T}$.

(d) Advantage of the observer that no fliter of this kind can give: the observer is driven by both the measurment y and the model with the actual input u, so it predicts the motion between samples.

The velocity estimate therefore has little or no phase lag. The low-pass fliter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c) = 8$ ms for its noice rejection, a trade-off it cannot escape. In addition the obsever gives all states, and can be augmented with a disturbance state (e.g. unknown load torque) which no fliter on θ can estimate.

Circumstance where the filter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or acuator saturation making u differ from the commanded value. The obsever propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free filter just smooths what the encoder actually measures.

Note: obsever poles are usually placed several times faster than the controller pols.

Question 4

Given: plant $\theta\ddot{} + a\theta\dot{} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\theta\dot{}$.

(a) Substitute the control law into the plant: $\theta\ddot{} + a\theta\dot{} + c\theta = b\,N_{ff}\,r - b\,k_1\,\theta - b\,k_2\,\theta\dot{}$. Moving the feedback terms to the left: $\theta\ddot{} + (a + b\,k_2)\,\theta\dot{} + (c + b\,k_1)\,\theta = b\,N_{ff}\,r$.

Numerically, $a + b\,k_2 = 2 + 2 = 4$ and $c + b\,k_1 = 6 + 2 = 8$, so the characterstic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stabel; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\,N_{ff} / (s^2 + 4s + 8)$. DC gain ($s = 0$): $T(0) = N_{ff}/8$. For unity DC gain, $\mathbf{N_{ff} = 8}$. (Note that this is larger than 1 because the gravity-related stiffness c and the feedback k_1 both resist the displacement.)

(b) With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\theta\ddot{} + (a + bk_2)\theta\dot{} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\theta\ddot{} + 3.5\theta\dot{} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 = \mathbf{0.8}$.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 = \mathbf{20\%}$. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, becuase b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The pols also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stabel.

(c) Substituting u and differentiating: the integral term enters with a minus sign, so the characteristic polynomal is $s^3 + (a + bk_2)s^2 + (c + bk_1)s - bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

So $2 + k_2 = 8 \Rightarrow k_2 = 6$; $6 + k_1 = 24 \Rightarrow k_1 = 18$; $-K_I = 32 \Rightarrow \mathbf{K_I = -32}$.

(d) Steady state: $\xi\dot{} = 0$, so $r = \theta_{ss}$; the parameters b and c do not enter, hence zero error for any b and c as long as it is stable.

Failure mode: windup when the actuator saturates. Remedy: reduce K_I to zero (or use a smaller gain) while saturated.